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F78AB
Actuarial and Financial Mathematics

Chapter 5: Real interest rates

LEADERS IN IDEAS AND SOLUTIONS

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Inflation…

- What is inflation?
- How do we measure inflation?
- Why is it important to measure inflation?
- How will inflation affect our decision making?

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Consider the price of sugar – 1 kg at the start of each month

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2000	166.6	167.5	168.4	170.1	170.7	171.1	170.5	170.5	171.7	171.6	172.1	172.2
2001	171.1	172.0	172.2	173.1	174.2	174.4	173.3	174.0	174.6	174.3	173.6	173.4
2002	173.3	173.8	174.5	175.7	176.2	176.2	175.9	176.4	177.6	177.9	178.2	178.5
2003	178.4	179.3	179.9	181.2	181.5	181.3	181.3	181.6	182.5	182.6	182.7	183.5
2004	183.1	183.8	184.6	185.7	186.5	186.8	186.8	187.4	188.1	188.6	189.0	189.9
2005	188.9	189.6	190.5	191.6	192.0	192.2	192.2	192.6	193.1	193.3	193.6	194.1

Table 7.1: The changing price in pence of 1kg of sugar.

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Consider the change in price of 1kg of sugar from January 2000 to January 2005

$$\frac{\text{Price of 1kg of sugar in Jan 2005}}{\text{Price of 1kg of sugar in Jan 2000}} = \frac{188.9}{166.6} = 1.13385\,,$$

i.e., the price of 1kg of sugar increased by 13.385% over 5 years

Expressed as an annual effective rate, the price change is

$(1.13385)^{1/5} - 1 = 0.02544,$

i.e., the annualised average change in the price of 1kg of sugar is +2.544% p.a. over 5 years.

Price of sugar has increased so purchasing power has gone down (you need more money to buy the same amount of sugar)

Accumulated average change in the price of a good

$$\% \text{ Change} = \frac{\text{Price of the good } n \text{ years later}}{\text{Price of the good in the base year}}$$

Price inflation is the change of purchasing power of money

Measured with reference to an index.

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Price inflation indices

- Retail Price Index (RPI)
- Consumer Price Index (CPI)

Measure the change in prices of a specified set (basket) of goods and services

Published by Office for National Statistics (ONS)

Other countries use similar measures

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Consumer Price Index (CPI)

- ONS looks at 180,000 prices of 743 items in the ‘shopping basket’
- Items change as spending habits change to keep up-to-date
- Index uses geometric average of prices

In 2025:

Added: VR headsets, men's sliders, exercise mats, pre-cooked pulled pork, and fixed tariffs for domestic gas and electricity

Removed: oven-ready joints, newspaper adverts and in-store cafeteria meals.

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Retail Price Index (RPI)

- Includes mortgage interest payments => heavily influenced by interest rates
- Arithmetic averaging

- Generally higher than CPI

- 12 months to December 2025: RPI 4.2%, CPI 3.4%

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Impact of inflation on yields

- To date we have calculated ***money yields*** – using the money payments to determine the rate of return that has been obtained.
- Over time, the money earned has lost purchasing power because of inflation
- We now calculate the ***real yield*** – the yield calculated using the real value of payments

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Real yields

- Determine *real value* of payments with reference to a chosen fixed date (the *base date)*

- Real yield does not depend on which base date we use

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Example

- Consider a two-year investment of £1,000, which returns £40 at the end of each of the next two years, along with return of the capital after the investment matures.

Calculate the effective return on the investment in money terms.

Now, given the following information about the inflation index

Time	0	1	2
Index	105	108	112

Calculate the real rate of return on the investment

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What does this tell us?

- Money yield = 4% p.a.
- Real yield = 0.7% p.a.
- What is the relationship between the money yield and the real yield?